5 In this question you must show detailed reasoning.

(a) Show that the gradient of the curve $y = \sqrt{x}\left(\frac{1}{x^2} - 2x\right)$ at the point $\left(\frac{1}{4}, \frac{31}{4}\right)$ is $-\frac{99}{2}$. **[4]**

(b) Find the equation of the tangent to the curve at $\left(\frac{1}{4}, \frac{31}{4}\right)$ giving your answer in the form $ax + by + c = 0$, where a, b and c are integers. **[2]**